

	1	2	3	4	5	6	7	8	
A	Integrated draft: power chain (sheets 2–4) and board stage (sheets 5–8). See DESIGN_SPEC.md. Inputs: VBAT 6–25.2 V (2–6S) from the MicoAir AM32 ESC pad row J3, and USB VBUS. Rails: 5V_IN from the U26 AP63205 buck, V5_SYS from the U25 TPS2121 mux, V3V3_SYS from the TPS62913 buck, V3V3_ANA from the TPS7A20 LDO. Servo 5 V is 5V_IN, upstream of the mux. No PCB placement or routing has been performed.								A
B	POWER 1 / ESC pad row, VBAT buck, USB, priority power mux File: power_sources.kicad_sch				BOARD 2 / IMU, barometer, high-g File: sensors.kicad_sch				B
C	POWER 2 / 3.3 V system buck and analog LDO File: power_3v3.kicad_sch				BOARD 3 / log flash and microSD File: storage.kicad_sch				C
D	POWER 3 / IO array File: power_periph.kicad_sch								D
E	BOARD 1 / RP2354B File: mcu.kicad_sch								E
F	1	2	3	4	5	6	7	8	

Sheet: /
File: MARV-V2.kicad_sch

Title: MARV integrated power – preliminary

Size: A3

Date: 2026-09-14

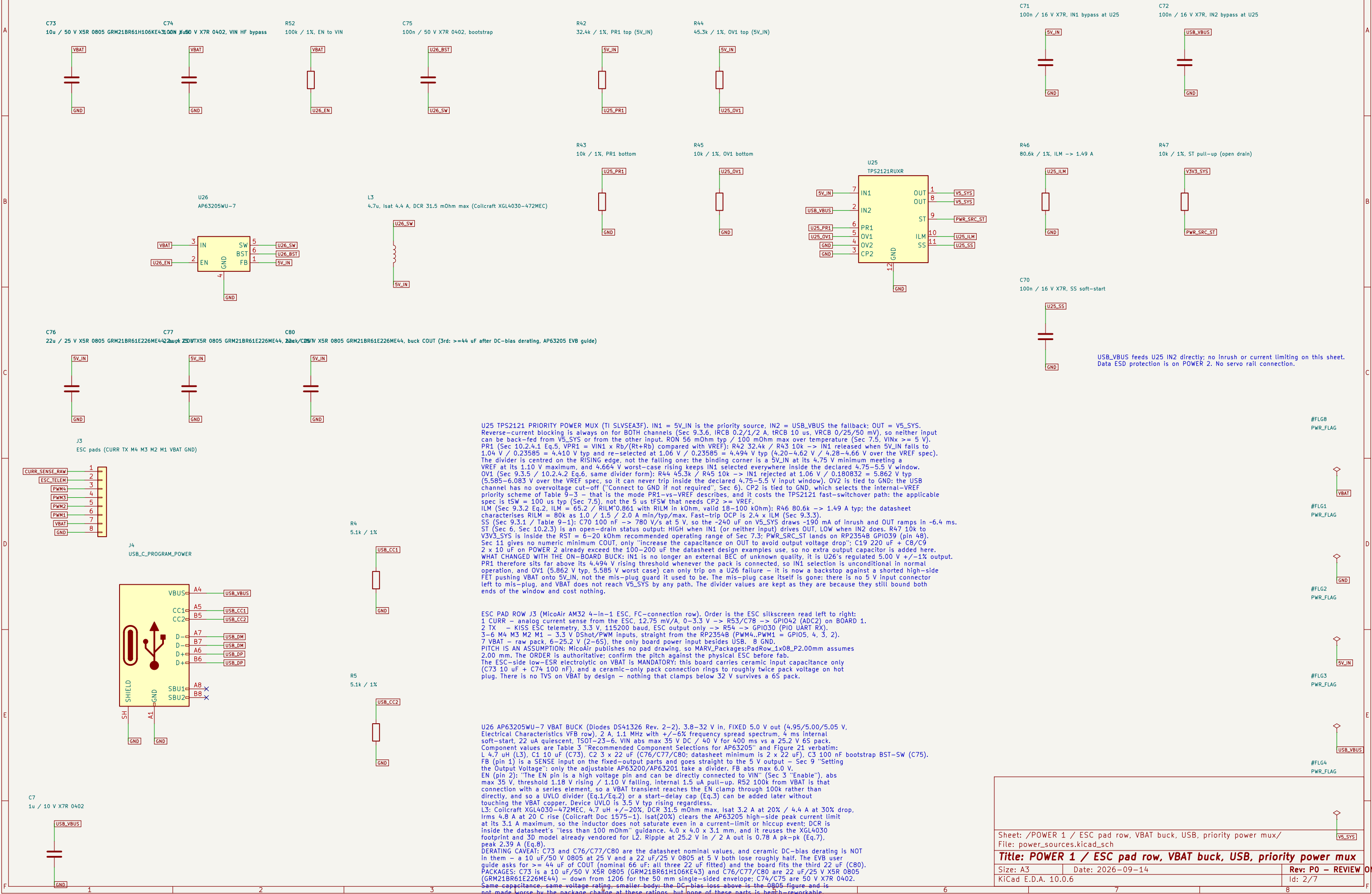
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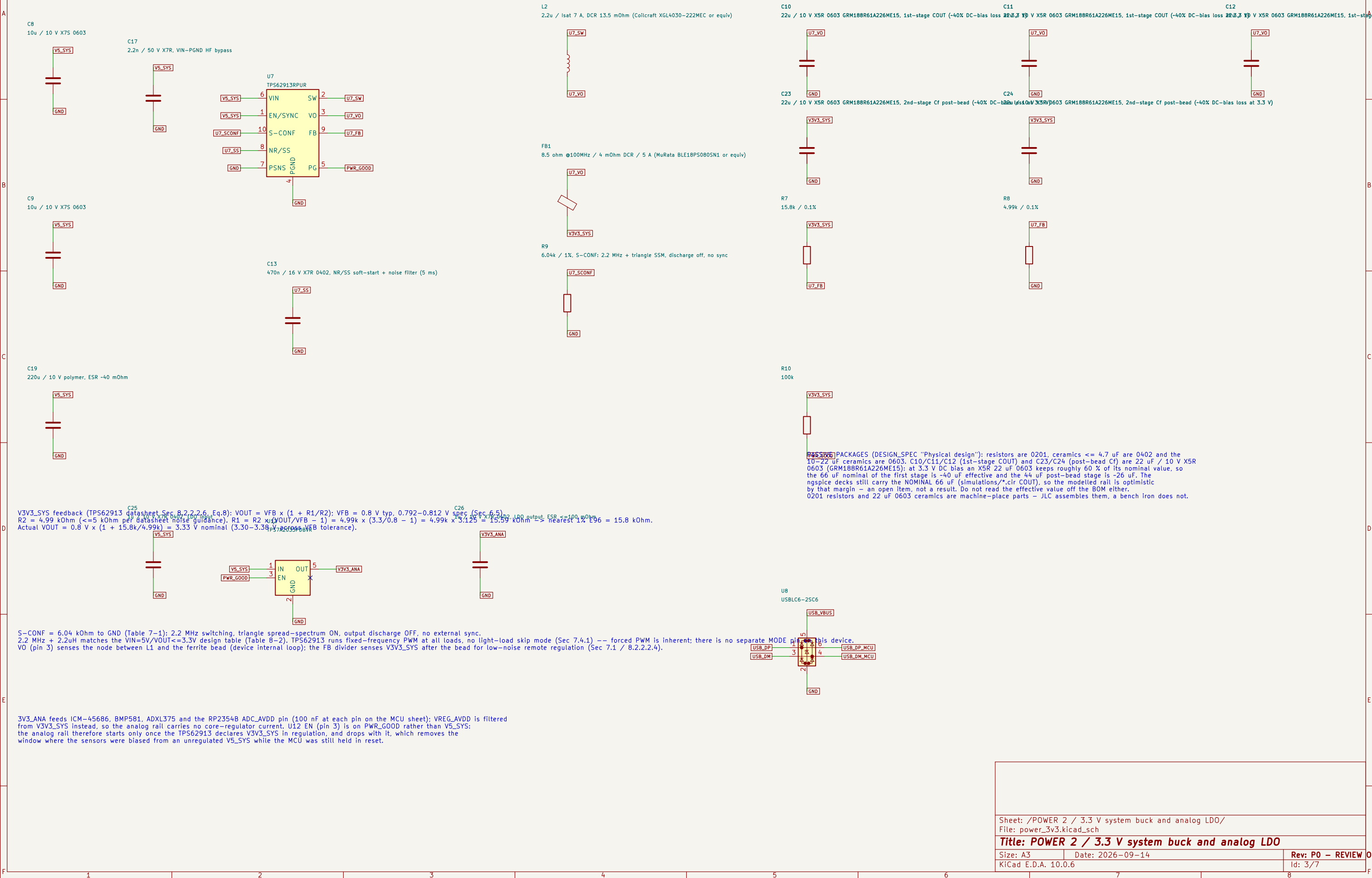
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Size: A3	Date: 2026-09-14	Rev: P0 – NOT FOR FAB
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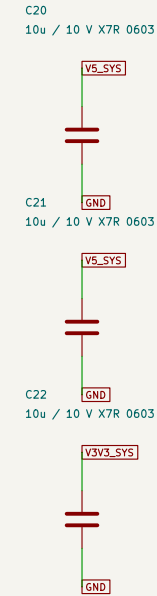
BOARD INPUTS: VBAT 6–25.2 V (2–6S) from the ESC pad row J3, and USB VBUS ~5 V (J4). U26 (AP63205WU) makes 5V_IN from VBAT; 5V_IN is the priority input of the U25 power mux and also the servo 5 V bus.



AVIONICS ONLY: 300 mA continuous / 500 mA short peak, provisional. NO SERVO POWER.

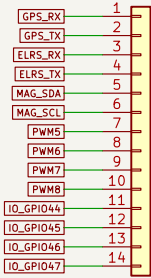


IO ARRAY – ONE 3-COLUMN x 14-ROW THROUGH-HOLE BLOCK ON THE RIGHT EDGE, 2.54 mm, madflitght FC3v2 style. It replaces the old 2x16 left-edge array AND the J12/J13/J14 servo block: every signal that leaves this board except the ESC row (J3) and the SWD pads (J10) now leaves on this one block. COLUMN RULE, from the board edge inward: GND (J8) | POWER (J7) | SIGNAL (J6). The SIGNAL column is the INNERMOST one, next to the MCU, so the 14 signal traces are the short ones; the two rails run down the outside where a plane stitch is cheap. A 3-wire module lead (signal, power, ground) plugs straight across ONE row. THREE Conn_01x14 SYMBOLS, not one 3x14: no 3-row generic connector symbol ships with this KiCad install, and three 1x14 rows with a body-trimmed courtyard abut exactly on the 2.54 mm grid (see tools/setup_pcb.py). DS-009 NAMING IS KEPT: a port label is named for the PERIPHERAL pin it belongs to, so GPS_RX / ELRS_RX are module inputs driven by the RP2354B UART TX pins (rows 1 and 3, silk T0 / T1) and GPS_TX / ELRS_TX are module outputs that land on the MCU UART RX pins (rows 2 and 4, silk R0 / R1).

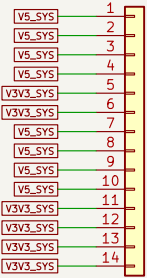


C20 / C21: V5_SYS local bulk at the array (rows 1–4 GPS/ELRS and rows 7–10 servo, the 5 V supply rows). C22: V3V3_SYS local bulk at the array (rows 5–6 and 11–14).

J6
IO array signal column (innermost)



J7
IO array power column (5V / 3V3 per row)



J8
IO array GND column (board edge)



IO ARRAY ROW MAP (row n = J6 pin n signal J7 pin n power J8 pin n GND)			
1	T0	GPS_RX	V5_SYS
2	R0	GPS_TX	V5_SYS
3	T1	ELRS_RX	V5_SYS
4	R1	ELRS_TX	V5_SYS
5	SDA	MAG_SDA	V3V3_SYS
6	SCL	MAG_SCL	V3V3_SYS
7	S5	PWM5	V5_SYS
8	S6	PWM6	V5_SYS
9	S7	PWM7	V5_SYS
10	S8	PWM8	V5_SYS
11	A44	IO_GPI044	V3V3_SYS
12	A45	IO_GPI045	V3V3_SYS
13	A46	IO_GPI046	V3V3_SYS
14	A47	IO_GPI047	V3V3_SYS

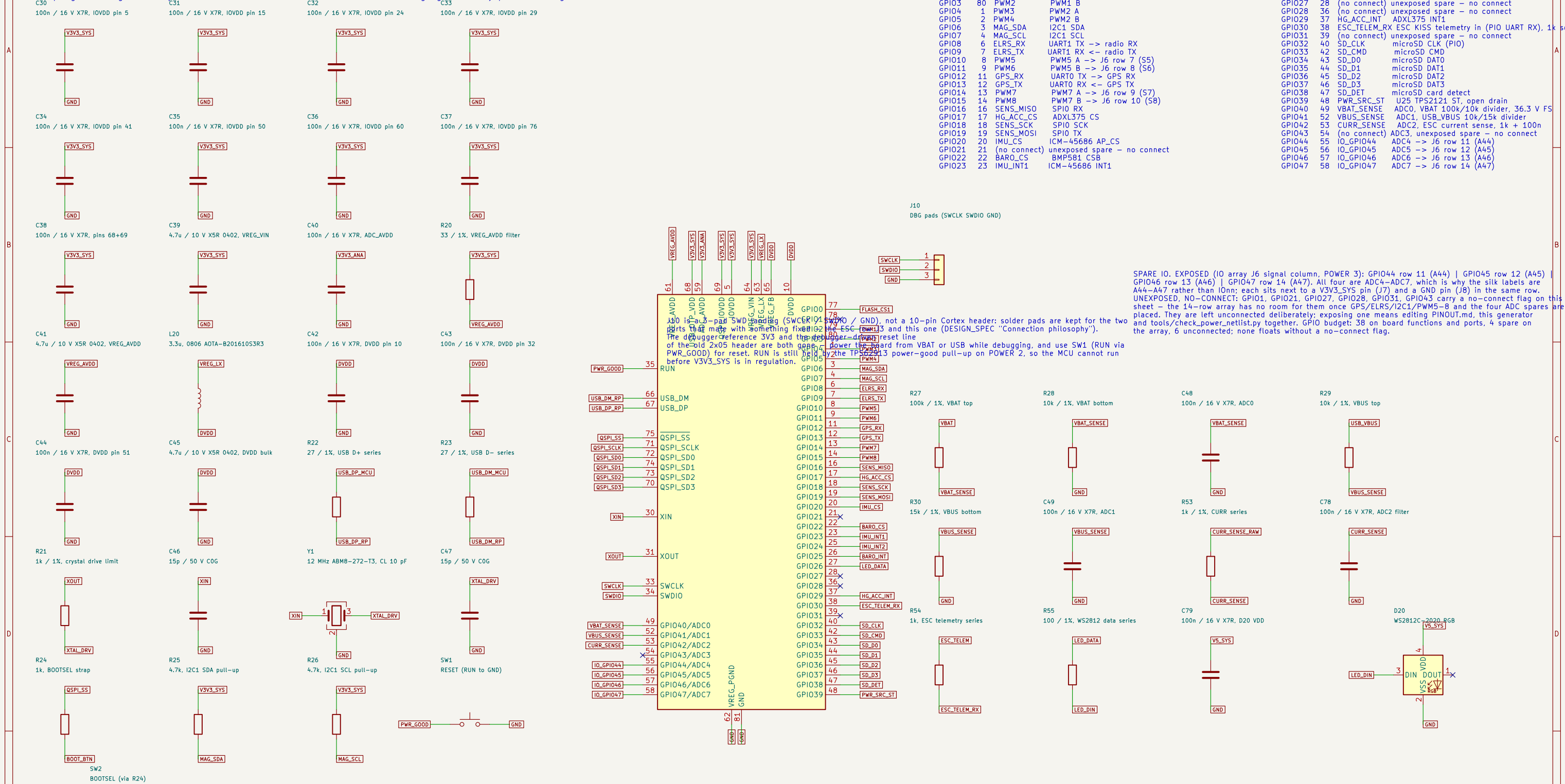
PWM MAP
PWM5 = GPIO10 (row 7, S5)
PWM6 = GPIO11 (row 8, S6)
PWM7 = GPIO14 (row 9, S7)
PWM8 = GPIO15 (row 10, S8)

ON THE ESC PAD ROW (J3, POWER 1)
PWM1 = GPIO2
PWM2 = GPIO3
PWM3 = GPIO4
PWM4 = GPIO5

RAILS ON THE ARRAY: rows 1–4 (GPS, ELRS) and rows 7–10 (servos S5–S8) take V5_SYS; rows 5–6 (magnetometer) and rows 11–14 (the four exposed spares) take V3V3_SYS. Every row has its own GND pin in the J8 column, so a signal return is never more than 2.54 mm away and no row can be mis-plugged into a neighbour's ground. SERVO 5 V NOW COMES FROM V5_SYS, NOT 5V_IN. The old J13 servo row tapped 5V_IN directly at the U26 buck output, ahead of the U25 priority mux, so servo current was outside the mux current limit. With one power column the array cannot carry two different 5 V nets under one "5V" silk label, so all eight 5 V rows are V5_SYS: servo current now crosses U25 and counts against its ILM limit (1.49 A, R46) and against the U26 buck. Budget servo current against BOTH – see DESIGN_SPEC "Rails and load budget". CURRENT: rows 1–4 and 7–10 count against the U26 buck through the mux; rows 5–6 and 11–14 count in the V3V3_SYS budget. No fuse and no per-row current limit: the array is a system-integration connector, not a protected port.

RP2354B, QFN-80, 2 MB in-package QSPI flash. Digital rails on V3V3_SYS, core DVDD from the on-chip buck.

RAIL SPLIT: V3V3_ANA (TPS7A20 LDO) now feeds exactly one RP2354B pin – ADC_AVDD (pin 59, with C40). The core regulator input filter R20/C41 that produces VREG_AVDD is fed from V3V3_SYS instead, so the buck-converter reference current no longer flows out of the low-noise analog rail and the sensors do not share a rail with it; V3V3_ANA is left carrying only ADC_AVDD and the MEMS. Decoupling, core regulator, crystal and BOOTSEL follow the RP2350 hardware design guide; every pin net is a global label.



PORT NAMING: a port label is named for the peripheral pin it comes from so GPS_TX / ELRS_TX are module outputs and land on MCU UART TX pins, while GPS_RX / ELRS_RX are module inputs fed by the MCU UART TX pins. On the 0 array (POWER 3) the UART rows are therefore 1 GPS_RX (silk T0) / 2 GPS_TX (R0) / 3 ELRS_RX (T1) / 4 ELRS_TX (R1). rows 5/6 are MAG_SDA / MAG_SCL, rows 7-10 the PWM5-8 servo signals and rows 11-14 the four exposed spares. The array is three stacked 1x14 rows: J6 SIGNAL, J7 POWER, J8 GND. All array SIGNAL pins are 3.3 V CMOS. GPIO39 (pin 48) is no longer spare: it reads PWR_SRC_ST, the open-drain ST output of U25 (TPS2121), which is high while 5V_IN (IN1) powers V5_SYS and low while USB VBUS (IN2) does.

Crystal circuit per design guide Sec 4: ABM8-272-T3, 15 pF each side (7.5 pF series + ~3 pF stray = 10.5 pF vs CL 10 pF), PWR_FLAG
 1 k in the XOUT leg to limit drive with a 50 ohm ESR crystal. C38 is the single design decoupling exception (pins 68+69 share it).
 VREG_AVDD RC = 33 ohm + 4.7 uF (design guide Sec 2.1), taken from V3V3_SYS; VREG_FB is tied to DVDD, VREG_PGND to GND, EP to GND.

TELEMETRY AND ANALOG INPUTS (100 nF at each ADC pin).
VBAT_SENSE: R27/R28 100k/10k scale VBAT by 1/11, so the 3.3 V ADC full scale is 36.3 V. A 6S pack at its 25.2 V maximum reads 2.29 V and a 2S pack at 6.0 V reads 0.55 V, so the whole 2–6S window is on-scale with headroom; the divider draws 229 μ A at 25.2 V. The old 5V/1N divider is gone with the 5 V input itself.
CURR_SENSE: the ESC current-sense output (J3 pin 1, 12.75 mV/A, 0–3.3 V) through R53 1k into C78 100 nF at the pin – a 100 μ S RC anti-alias/ESD network, well under the RP2350 ADC source-impedance guidance.
ESC_TELEM: the ESC KISS telemetry output (J3 pin 2, 3.3 V, 115200 baud, one-wire from the ESC) through R54 1k into GPIO30, read by a PIO UART. The resistor is series protection only, not a level shifter: the ESC pad row is 3.3 V CMOS on both the telemetry and the four motor lines.
VBUS_SENSE keeps 10k/15k on GPIO41.
DOUT_WS2812C–2020: VDD on V5_SVS with C79 100 nF, DIN from GPIO26 through R55 100 R, DOUT no-connect (single pixel). Its DIN VIH minimum is 2.7 V (datasheet Electrical Characteristics), which 3.3 V CMOS clears – that is why the C variant is used rather than a 5050 WS2812B, whose VIH is 0.7 $\times$ VDD = 3.5 V.
QSPI* are the dedicated QSPI pads: they reach the in-package flash die and the package pins (RP2350 datasheet Sec 14.3), so U24 shares the bus with GPIO0/QM1 CS1n (FLASH_CS1) as its select. RP2354 requires QSPI_OVDD = 3.3 V, and IOVDD = 3.3 V to run a second QSPI device (Sec 14.9).

PORT NAMING: a port label is named for the peripheral pin it comes from, so GPS_TX / ELRS_TX are module outputs and land on MCU UART RX pins, while GPS_RX / ELRS_RX are module inputs fed by the MCU UART TX pins. On the 0 array (POWER 3) the UART rows are therefore 1 GPS_RX (silk T0) / 2 GPS_TX (R0) / 3 ELRS_RX (T1) / 4 ELRS_TX (R1). rows 5/6 are MAG_SDA / MAG_SCL, rows 7-10 the PWM5-8 servo signals and rows 11-14 the four exposed spares. The array is three stacked 1x14 rows: J6 SIGNAL, J7 POWER, J8 GND. All array SIGNAL pins are 3.3 V CMOS. GPIO39 (pin 48) is no longer spare: it reads PWR_SPSST, the open-drain ST output of U25 (TPS2121), which is high when V5_IN (IN1) powers V5_SYS and low while USB VBUS (IN2) does.

```

PWR_FLAG
share it)
PWR_FLAG
PWR_FLAG

```

EP to GND.

• **Prevalence** = the proportion of a population that has a disease at a particular point in time

MECHANICAL: H1-H4 are the four 4.0 mm NPTH grogmet holes on the standard 30.5 x 30.5 mm pattern, centred on the board, each with a 5.0 mm copper keepout and an 8.0 mm top-side courtyard for the grogmet flange. No pins, no nets. See DESIGN_SPEC "Physical design".

Sheet: /BOARD 1 / RP2354B/

File: mcu.kicad_sch

Title: BOARD 1 / RP2354B

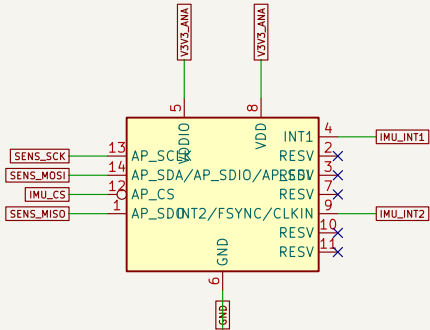
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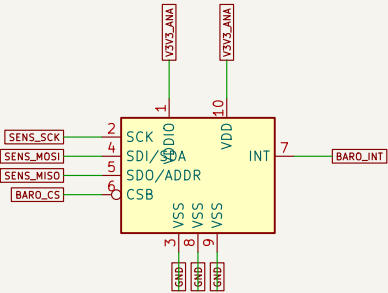
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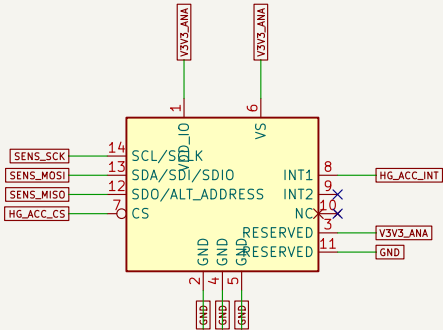
All three sensors run 4-wire SPI on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus with one chip select each, and are powered from V3V3_ANA (TP57A20 LDO). Every supply pin on this sheet gets 100 nF + 1 uF; BMP581 and ADXL375 because their own datasheets call for the pair, the ICM-45686 as a deliberate addition – its datasheet BOM (DS-000577 Rev 1.0 Table 11) lists 0.1 uF only.



U21
ICM-45686 6-axis IMU



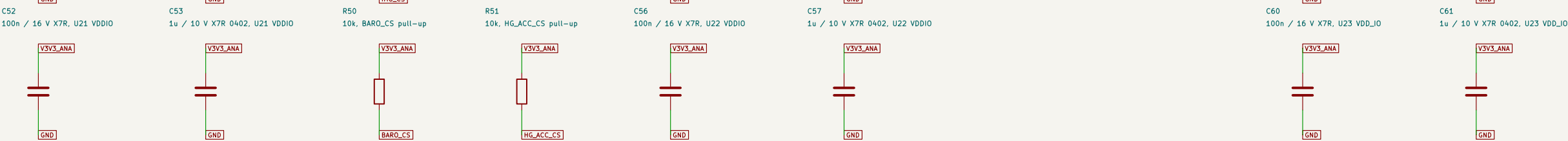
U22
BMP581 barometer



U23
ADXL375 high-g accelerometer



TEST POINTS TP1-TP10 (TestPoint:TestPoint_Pad_1.0x1.6mm, 1 x 1 mm bare F.Cu pads, 50 model by design – tools/audit footprints by --no-3d-ok covers TestPoint, along with the pad rows and mounting holes). TP1 SENS_SCK | TP2 SENS_MOSI | TP3 SENS_MISO | TP4 IMU_CS | TP5 BARO_CS | TP6 HG_ACC_CS | TP7 IMU_INT1 | TP8 IMU_INT2 | TP9 BARO_INT | TP10 HG_ACC_INT. Placement: Inside the sensor cluster, within 6 mm of U21/U22/U23 (tools/setup_pcb.py checks it). They add ten stubs to the SPI bus, which is why they are 1 mm pads next to the parts rather than a header somewhere else: at 10 MHz SCK a short stub is a capacitive load, a long one is a transmission line. No series resistors and no ground pad of their own – probe ground comes off a mounting hole or the J6 array.



U21 ICM-45686 (TDK InvenSense DS-000577 Rev 1.0, Table 10 "Signal Descriptions" / Figure 4 pin-out, Single Interface SPI mode, Figure 10): AP_CS (pin 12) = IMU_CS, AP_SCLK (13) = SENS_SCK, AP_SDA/AP_SDIO/AP_SDI (14) = SENS_MOSI, AP_SDO/AP_ADO (1) = SENS_MISO, INT1 (4) = IMU_INT1, INT2/FSYNC/CLKIN (9) = IMU_INT2 – typed bidirectional because FSNC/CLKIN are host-driven inputs while INT2 is a chip output; only the INT2 function is used here. VDD (8) and VDDIO (5) are both on V3V3_ANA; GND (6) is the only ground pin. Pins 2, 3, 7, 10 and 11 (RESV) are all left No Connect; Table 10 allows No Connect / VDDIO / GND for every RESV pin, and the Single-Interface SPI typical application schematic (Figure 10) shows all five left unterminated (internal pull enabled by default, per the per-pin notes in Table 10); they are only needed for the AUX1 or I2C-master dual-interface modes, neither of which this board uses. C50 (VDD) and C52 (VDDIO) are the 100 nF X7R of Table 11 "Bill of Materials" (C1/C2 in the datasheet); C51 (VDD) and C53 (VDDIO) add 1 uF on top of them. The datasheet BOM lists 0.1 uF only; this board keeps 100 nF + 1 uF per supply pin for consistency with U22/U23 and common flight-controller practice.

U22 BMP581: CSB = BARO_CS, SDI = SENS_MOSI, SDO = SENS_MISO, SCK = SENS_SCK, INT = BARO_INT, VDDIO and VDD on V3V3_ANA, all three VSS pins to GND.
U23 ADXL375: CS = HG_ACC_CS, SDA/SDI = SENS_MOSI, SDO = SENS_MISO, SCL/SCLK = SENS_SCK, INT1 = HG_ACC_INT. Per the pin function table, pin 3 (RESERVED) goes to VS and pin 11 (RESERVED) to GND; pin 10 (NC) and the unused INT2 are left open. VS and VDD I/O are both on V3V3_ANA.

R48, R50, R51: 10k pull-ups to V3V3_ANA on the three SPI chip selects (IMU_CS, BARO_CS, HG_ACC_CS). R49 (the former second IMU chip-select pull-up) is removed along with that net – the ICM-45686 has a single chip select. Same rationale as R35 on FLASH_CS1 (BOARD 3): every sensor is held deselected before the RP2354B drives the pin, so a GPIO left floating through reset, BOOTSEL or a debugger halt cannot open a transaction on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus. They bias to V3V3_ANA, the same rail as every VDDIO pin on this sheet, so there is no pull-up current path into an unpowered I/O supply and no CS pin is pulled above its own VDDIO during rail sequencing.

Sheet: /BOARD 2 / IMU, barometer, high-g/
File: sensors.kicad_sch

Title: BOARD 2 / IMU, barometer, high-g

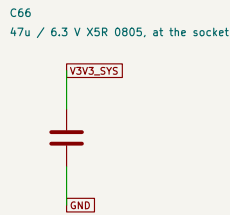
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Sheet: /BOARD 3 / log flash and microSD/
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Date: 2026-09-14

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